

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

6 Further Graphics (pb355)

Basis Splines

(a) Consider the b-spline curve $P(t)$ with knot vector $[0, 1, 2, 3, 3, 3]$ and $k = 3$.

(i) In a single sentence, explain the effect on $P(t)$ of repeating a knot value k times. [1 mark]

Answer:

All questions on this topic are from Lecture 5.

Repeating a knot value k times causes the curve to pass directly through the corresponding control point.

(ii) State the equations that define $P(t)$. [3 marks]

Answer:

Marking scheme: full points for the right answer, or,

- 1 point for remembering that the definition is recursive
- 1 point for the base case
- 1 point for the recursive case

From lectures notes slide 90:

$$P(t) = \sum_{i=1}^n N_{i,k}(t)P_i, \quad t_{\min} \leq t < t_{\max}$$

$$N_{i,1}(t) = \begin{cases} 1, & t_i \leq t < t_{i+1} \\ 0, & \text{otherwise} \end{cases}$$

$$N_{i,k}(t) = \frac{t-t_i}{t_{i+k-1}-t_i}N_{i,k-1}(t) + \frac{t_{i+k}-t}{t_{i+k}-t_{i+1}}N_{i+1,k-1}(t)$$

(iii) State the equation and sketch the graph for each of the three quadratic basis functions $N_{i,3}(t)$ of $P(t)$. [8 marks]

Answer:

Marking scheme: full points for all six answers being correct without showing work; or, if all answers are not correct,

- 1 point for expressing that each of the equations is assembled from multiple polynomials, conditional on t
- 3 points for accurate equations (1 apiece)
- 2 points for following the correct steps to derive the equation expressions (see below), even if algebraic error led to the wrong outcome
- 1 point if the graphs are approximately correct (see below)
- 1 point if the graphs are bounded in the correct range of t (beginning at the $t = 0, 1$, and 2 respectively; ending at $t = 3$) and all sketched to be less than 1.0 .

For parameter $k = 1$:

$$N_{1,1}(t) = 1, 0 \leq t < 1 \quad N_{4,1}(t) = 1, 3 \leq t < 3$$

$$N_{2,1}(t) = 1, 1 \leq t < 2 \quad N_{5,1}(t) = 1, 3 \leq t < 3$$

$$N_{3,1}(t) = 1, 2 \leq t < 3$$

— *Solution notes* —

For parameter $k = 2$:

$$N_{1,2}(t) = \frac{t-0}{1-0}N_{1,1}(t) + \frac{2-t}{2-1}N_{2,1}(t) = \begin{cases} t & 0 \leq t < 1 \\ 2-t & 1 \leq t < 2 \end{cases}$$

$$N_{2,2}(t) = \frac{t-1}{2-1}N_{2,1}(t) + \frac{3-t}{3-2}N_{3,1}(t) = \begin{cases} t-1 & 1 \leq t < 2 \\ 3-t & 2 \leq t < 3 \end{cases}$$

$$N_{3,2}(t) = \frac{t-2}{3-2}N_{3,1}(t) + \frac{3-t}{3-3}N_{4,1}(t) = \begin{cases} t-2 & 2 \leq t < 3 \\ \dots & 3 \leq t < 3 \end{cases}$$

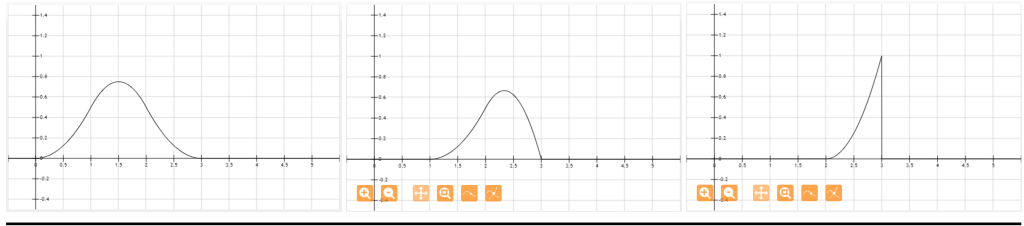
$$N_{4,2}(t) = \frac{t-3}{3-3}N_{4,1}(t) + \frac{3-t}{3-3}N_{5,1}(t) = \begin{cases} \dots & 3 \leq t < 3 \\ \dots & 3 \leq t < 3 \end{cases}$$

For parameter $k = 3$:

$$N_{1,3}(t) = \frac{t-0}{2-0}N_{1,2}(t) + \frac{3-t}{3-1}N_{2,2}(t) = \begin{cases} \frac{t^2}{2} & 0 \leq t < 1 \\ -t^2 + 3t - \frac{3}{2} & 1 \leq t < 2 \\ \frac{t^2}{2} - 3t + \frac{9}{2} & 2 \leq t < 3 \end{cases}$$

$$N_{2,3}(t) = \frac{t-1}{3-1}N_{2,2}(t) + \frac{3-t}{3-2}N_{3,2}(t) = \begin{cases} \frac{t^2}{2} - t + \frac{1}{2} & 1 \leq t < 2 \\ \frac{-3t^2}{2} + 7t - \frac{15}{2} & 2 \leq t < 3 \\ \dots & 3 \leq t < 3 \end{cases}$$

$$N_{3,3}(t) = \frac{t-2}{3-2}N_{3,2}(t) + \frac{3-t}{3-3}N_{4,2}(t) = \begin{cases} t^2 - 4t + 4 & 2 \leq t < 3 \\ \dots & 3 \leq t < 3 \\ \dots & 3 \leq t < 3 \end{cases}$$



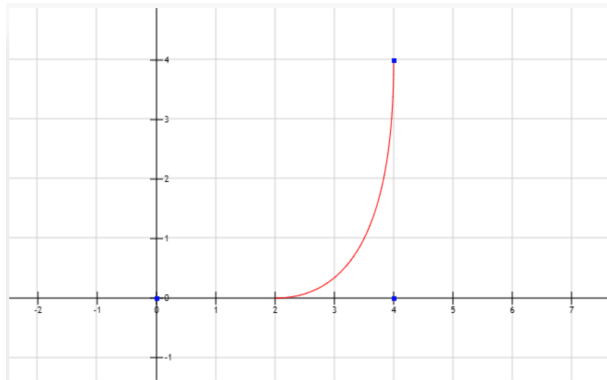
(iv) Plot the path of $P(t)$ for control points $P_0 = (0, 0)$, $P_1 = (4, 0)$, $P_2 = (4, 4)$.
[4 marks]

Answer:

The curve traces an upwards-curving hyperbola from $(2, 0)$ to $(4, 4)$.

Marking scheme: full points for the right answer, or,

- 1 point if the curve is defined from $t = 2$ to $t = 3$
- 1 point if the curve has one endpoint at $(2, 0)$ and the other at $(4, 4)$
- 1 point if the curve at least attempts a simple quadratic



Computational
Geometry

- (b) Consider an embedded closed manifold surface with 48 vertices, 48 faces, and 100 edges.

- (i) What is the genus of this surface, and what is the formula to find it?
[2 marks]

Answer: The Poincaré Formula states that $V - E + F = 2 - 2g = \chi$.

$$48 - 100 + 48 = -4 = 2 - 2g \rightarrow g = 3$$

- (ii) What is the total angle deficit of this surface, and what is the formula to find it?
[2 marks]

Answer: Descartes' Theorem of Total Angle Deficit states that on a surface S with Euler characteristic χ , the sum of the angle deficits of the vertices is $2\pi\chi$.

$$\chi = -4 \rightarrow AD = -8\pi$$
